



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology(FST)

Software Engineering

**AIUB Portal Redefined: Integrating Campus Life with
Smart Digital Innovation**

Submitted To

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Project Process Model

Submitted By

Semester: Summer 2024-2025		Section: D	Group Number: 01	
SN	Student Name	Student ID	Contribution (CO3)	Individual Marks
01	NILOY PAUL	23-51773-2		
02	ARNOB SARKER SUPTA	23-52080-2		
03	ABDULLAH AL TAMJID	23-51887-2		
04	SADIA ISLAM	23-51943-2		
05	KAHIUM AHAMED FAHIM	22-48593-3		

The project will be Evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	

Rubric for Project Assessment (CO3)

Marks distribution (Max 3X5= 15)					
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present Confusing organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	Some errors in spelling and grammar. Some problems of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	Few errors in spelling and grammar. Presents most of the details in a logical flow of organization in definition, details, and example.	Project report is complete and No errors in spelling and grammar. Consistently presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Description of Student's Contribution in the Project work

Student Name: NILOY PAUL Student ID: 23-51773-2 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Objective ▪ Project Role Identification and Responsibilities <u>NILOY PAUL</u> Signature of the Student
Student Name: ARNOB SARKER SUPTA Student ID: 23-52080-2 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Selected Process Model: SCRUM ▪ Evidence Supporting SCRUM Selection for AIUB Portal Redefined <u>ARNOB SARKER SUPTA</u> Signature of the Student
Student Name: ABDULLAH AL TAMJID Student ID: 23-51887-2 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Arguments and Reasons for SCRUM Selection <u>ABDULLAH AL TAMJID</u> Signature of the Student
Student Name: SADIA ISLAM Student ID: 23-51943-2 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Impact Analysis on Societal, Health, Safety, Legal, and Cultural Issues <u>SADIA ISLAM</u> Signature of the Student
Student Name: KAHIUM AHAMED FAHIM Student ID: 22-48593-3 Contribution in Percentage (20%): <u>Contribution in the Project:</u> ▪ Analysis of Software Nature and Environment <u>KAHIUM AHAMED FAHIM</u> Signature of the Student

SOFTWARE DEVELOPMENT LIFE CYCLE

Project Title

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation.

[1] Process Model

1.I. Objective

The enhanced AIUB student portal intends to enhance the entire student experience by integrating academic, administrative, and personal support services on a single platform. The new features include addressed by AIUB Portal Redefined

- CGPA & Demand-Based Course Registration System
- Lost and Found Submission
- Smart Location System (AI + AR Navigation)
- Event and Notice Display
- Gymnasium Registration & Payment
- Anonymous Complaint System
- Course Book Reference & Online Resources
- Global Printing System
- Faculty Information Directory
- AIUB File Sharing System
- Class Routine & Upcoming Registration Course Generator

1.II. Analysis of Software Nature and Environment

- **Project Nature:** The "AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation" is a complex, multi-module system integrating AI-driven course registration, AR-based navigation, cloud printing, anonymous complaints, and file sharing. It serves a diverse user base (students, faculty, administrators) in a dynamic university environment, requiring high interactivity and adaptability.
- **Environment Characteristics:**
 - **Dynamic Requirements:** Features like CGPA-based registration, AR navigation, and anonymous complaints may evolve based on user feedback and institutional needs.
 - **Technological Complexity:** Combines React.js, Node.js, MySQL, ARKit/ARCore, TensorFlow, AWS, and BLE beacons, necessitating iterative integration and testing.
 - **User-Centric Design:** Requires continuous feedback to ensure usability (e.g., intuitive navigation, accessible interfaces) and inclusivity (e.g., wheelchair-friendly routes).
 - **Scalability Needs:** Must handle peak loads (e.g., registration periods) with >99% uptime, as specified in the SRS non-functional requirements.
 - **Societal and Legal Considerations:** Anonymous complaints and data privacy (HTTPS, encryption) address safety and legal compliance, while accessibility modes support inclusivity.
 - **Budget and Timeline:** Operates within AIUB's academic budget and a 16-month timeline (SRS: 3+6+4+3 months), requiring cost-effective, modular development.

1.III. Selected Process Model: SCRUM

- **Rationale:** The existing SCRUM model (as per "SE - Project Process Model [OBE].docx") is well-suited due to its iterative, user-focused approach, which aligns with the project's dynamic requirements, complex technology stack, and need for continuous stakeholder engagement. However, the documentation can be enhanced with more precise arguments, evidence, and impact analysis to meet the evaluation criteria comprehensively.

1.IV. Arguments and Reasons for SCRUM Selection

- **Adaptability:** SCRUM's sprints enable rapid adjustments to evolving needs like course registration and AR navigation, unlike Waterfall's rigid phases.
- **Technology Integration:** Iterative sprints support parallel development and testing of complex modules (e.g., BLE beacons, payment gateways), unlike V-Model's sequential approach.
- **User-Centric:** Sprint reviews ensure usability and accessibility (e.g., <2-second response, wheelchair-friendly routes), unlike Spiral's risk focus.
- **Incremental Delivery:** Early module delivery (e.g., registration) aligns with SRS phased deployment, unlike Waterfall's delayed release.
- **Scalability:** Incremental releases test scalability for peak loads (>99% uptime), per SRS requirements.
- **Risk Mitigation:** Daily standups catch issues early (e.g., payment bugs), unlike Waterfall's late discovery.
- **AIUB Alignment:** SCRUM delivers a scalable, user-centric platform, meeting SRS performance goals and AIUB's digital transformation vision.

1.V. Evidence Supporting SCRUM Selection for AIUB Portal Redefined

The SCRUM model is the most suitable for the "AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation" project, as it aligns with the dynamic, complex, and user-centric nature of the system. Below is a detailed comparison with other process models, supported by specific evidence from the project proposal and SRS, focusing on features like CGPA & Demand-Based Course Registration System, Smart Location System (AI + AR Navigation), Global Printing System, Anonymous Complaint System, and others.

1. Compared to Waterfall:

- **Evidence:** The project proposal highlights the need for a flexible system to address evolving needs, such as CGPA-based course registration and AR navigation, which may change due to user feedback or new university policies (e.g., accessibility for disabled users). Waterfall's linear structure locks phases (e.g., planning, design), making changes costly and difficult. SCRUM's 2–3-week sprints, as outlined in the SRS implementation phases, enable adaptive planning, allowing adjustments to features like course registration waitlists or AR navigation prompts based on real-time feedback, ensuring alignment with AIUB's student-centric goals.

2. Compared to V-Model:

- **Evidence:** The V-Model's heavy focus on verification and validation per phase is time-consuming and suited for fixed requirements. The SRS specifies features like real-time schedule conflict detection in course registration and encrypted anonymous complaints, which require iterative user validation to ensure usability and security. SCRUM's sprint reviews, planned in the SRS pilot testing phase, facilitate continuous feedback from students and faculty, allowing features like event notices and gym payments to evolve with semester-based needs, unlike the V-Model's rigid structure.

3. Compared to Prototyping Model:

- **Evidence:** The Prototyping Model excels at visualizing ideas but lacks structure for large systems. The project proposal outlines 11 diverse features, including a searchable lost and found database and faculty directory, requiring scalability and maintainability. SCRUM's structured sprints and roles (e.g., Product Owner for backlog management), as per the SRS iterative development phase, ensure disciplined delivery of complex modules like AI-driven course demand analysis and AR overlays with BLE beacons, unlike the unstructured Prototyping Model.

4. Compared to Incremental Development:

- **Evidence:** Incremental Development delivers system parts over time but lacks SCRUM's collaborative framework. The SRS prioritizes features like course registration and global printing, requiring stakeholder feedback from students and admins. SCRUM's organized sprints, daily standups, and reviews, as detailed in the SRS pilot testing, enhance cooperation and refine features like the waitlist mechanism and cloud printing notifications, ensuring agility and progress, unlike Incremental Development's limited focus on collaboration.

5. Compared to Component-Based Development:

- **Evidence:** Component-Based Development emphasizes reusable components, ideal for modularity but less adaptive to changing needs. The project proposal highlights dynamic features like anonymous complaints and course resources, which need ongoing user-driven updates. SCRUM's iterative cycles, as per the SRS pilot testing phase, support continuous refinement of features like wheelchair-friendly navigation routes and secure file sharing links, aligning with AIUB's inclusivity goals, unlike Component-Based Development's static focus.

6. Compared to RAD (Rapid Application Development):

- **Evidence:** RAD focuses on quick prototyping with minimal documentation, suitable for small projects but inadequate for the complex system described in the proposal (e.g., 11 integrated features). The SRS requires scalability (>99% uptime) and security (HTTPS, encryption) for features like global printing and anonymous complaints. SCRUM's disciplined framework, with sprint reviews and backlog prioritization, ensures quality and scalability, as outlined in the SRS deployment phase, unlike RAD's lack of structure for multi-module systems.

7. Compared to RUP (Rational Unified Process):

- **Evidence:** RUP's process-heavy phases (e.g., Inception, Construction) suit large enterprise systems but are slow for academic contexts. The proposal emphasizes rapid adaptation for features like event notices and gymnasium registration. SCRUM's lightweight sprints, aligned with the SRS's 16-month timeline, support quick iterations and stakeholder feedback, ensuring features like course resource access and faculty directory updates meet AIUB's dynamic needs, unlike RUP's complex structure.

8. **Compared to Basic Agile:**

- **Evidence:** Basic Agile offers flexibility but lacks a defined framework. The SRS specifies structured development with roles (e.g., Product Owner, Scrum Master) and artifacts (e.g., product backlog), aligning with SCRUM's events like sprint planning. This ensures measurable progress for features like AI-driven course suggestions and AR navigation with voice guidance, making SCRUM more actionable than Basic Agile for AIUB's multi-stakeholder environment.

9. **Compared to XP (Extreme Programming):**

- **Evidence:** XP prioritizes coding practices, ideal for small teams but less effective for stakeholder coordination. The proposal requires collaboration for features like faculty directory and anonymous complaints. SCRUM's roles and sprint reviews, as per the SRS pilot testing, ensure stakeholder input from students, faculty, and admins, delivering a comprehensive system with features like secure file sharing and print usage reports, unlike XP's technical focus.

10. **Alignment with Project Goals and SRS Requirements:**

- **Evidence:** The proposal underscores AIUB's digital transformation vision, requiring a scalable, user-centric platform with features like CGPA-based registration, AR navigation, and cloud printing. SCRUM's iterative cycles align with the SRS's phased approach (Research, Development, Pilot Testing, Deployment), enabling early delivery of high-priority modules (e.g., registration, printing) while refining others (e.g., file sharing). The SRS's non-functional requirements (<2-second response, >99% uptime) are supported by SCRUM's continuous testing, ensuring performance and reliability across all features.

1.VI. **Impact Analysis on Societal, Health, Safety, Legal, and Cultural Issues**

• **Societal Impact:**

- **Equitable Education:** CGPA-based registration ensures fair course access, addressing social equity.
- **Inclusivity:** Accessibility modes in AR navigation support disabled users, promoting equal access.

• **Health Impact:**

- **Student Wellness:** Online gymnasium registration encourages fitness, enhancing student health.
- **Mental Health:** Streamlined registration and navigation reduce stress from manual processes.

• **Safety Impact:**

- **Anonymous Complaints:** Encrypted submissions encourage reporting of harassment, improving campus safety.
- **Navigation Safety:** AR navigation reduces confusion, minimizing risks in crowded campus areas.

• **Legal Impact:**

- **Data Privacy:** HTTPS, RBAC, and encryption (SRS non-functional requirements) ensure compliance with data protection laws.
- **Auditability:** Time-stamped registration and printing logs support legal transparency.

• **Cultural Impact:**

- **Community Engagement:** Centralized event notices foster participation in cultural and academic activities.
- **Ethical Sensitivity:** SCRUM's stakeholder feedback ensures alignment with AIUB's cultural values, such as inclusivity and transparency.

[2] Project Role Identification and Responsibilities

2.I. Identified Responsibilities of the Roles and Stakeholders

- **Product Owner:**
 - **Responsibilities:**
 - Gathers requirements from AIUB stakeholders (students, faculty, admins).
 - Prioritizes features (e.g., high-priority registration, printing; medium-priority navigation, complaints).
 - Defines user stories with acceptance criteria (e.g., “Students can register without conflicts”).
 - Ensures alignment with AIUB’s academic and administrative goals.
- **Scrum Master:**
 - **Responsibilities:**
 - Facilitates sprint planning, daily standups, reviews, and retrospectives.
 - Removes blockers (e.g., delays in AWS setup, stakeholder feedback delays).
 - Ensures adherence to SCRUM principles and timely delivery of increments.
 - Coaches team on Agile practices to maintain productivity and quality.
- **Software Developers:**
 - **Responsibilities:**
 - Develop frontend (React.js) for intuitive interfaces (e.g., registration forms, navigation maps).
 - Build backend (Node.js) and database (MySQL/SQL Server) for data management.
 - Integrate APIs for payment gateways (bKash) and file sharing.
- **AI/ML Engineers:**
 - **Responsibilities:**
 - Train AI models (TensorFlow, Scikit-learn) for course demand analysis and AR route optimization.
 - Ensure model accuracy (e.g., >95% for demand predictions, per SRS evaluation metrics).
- **AR Developers:**
 - **Responsibilities:**
 - Implement AR navigation using ARKit, ARCore, and Navignine SDK.
 - Integrate BLE beacons and QR code markers for precise indoor positioning (<2-second response).
- **UI/UX Designers:**
 - **Responsibilities:**
 - Design mobile-friendly, accessible interfaces for registration, navigation, and dashboards.
 - Ensure inclusivity with wheelchair-friendly navigation modes and intuitive layouts.
- **QA/Test Engineers:**
 - **Responsibilities:**
 - Conduct unit, integration, and user acceptance testing for all modules.
 - Validate AR navigation accuracy, registration logic, and payment security.

- **DevOps Engineers:**
 - **Responsibilities:**
 - Manage AWS deployment, CI/CD pipelines, and system monitoring.
 - Ensure scalability (e.g., handling peak registration loads) and >99% uptime.
- **System Administrators:**
 - **Responsibilities:**
 - Maintain security (HTTPS, RBAC, encryption for complaints).
 - Update databases (faculty directory, course resources) and generate print usage reports.
- **Customers/Stakeholders (Students, Faculty, Admins):**
 - **Responsibilities:**
 - Provide feedback during sprint reviews on usability (e.g., navigation ease, complaint system).
 - Validate features in pilot testing (SRS Phase 3) and suggest improvements.
 - Clarify requirements via the Product Owner (e.g., event notice filters, resource access).
- **Management (AIUB Leadership):**
 - **Responsibilities:**
 - Allocate budget, infrastructure, and resources (e.g., AWS servers, BLE beacons).
 - Align project with AIUB's strategic goals (digital transformation, student satisfaction).
 - Support stakeholder engagement and remove organizational barriers.